Features: 1.Based on ESP32-S3, dual core 240MHz, powerful AI computing power. 2. Supports Wi Fi+Bluetooth 5 (LE) for more stable wireless connection. 3. Isolated RS485/CAN interface for safer industrial communication. 4. Onboard RS485 and CAN indicator lights for easy sensing of device operation status. 5. The guide rail type shell design is easy to install and has strong protection. Product Features: Based on ESP32-S3 microcontroller, built-in Xtensa 32-bit LX7 dual core processor, with a clock speed of up to 240MHz Integrated 2.4GHz Wi Fi and low-power Bluetooth BT 5 (LE) dual-mode wireless communication, with excellent RF performance Onboard isolated RS485 interface, convenient for external expansion and access to various RS485 Modbus industrial expansion modules or sensors Onboard isolated CAN interface, convenient for external expansion and access to various CAN devices Onboard pin interface, allowing access to other devices Onboard USB Type-C interface, can be used for power supply and download debugging, convenient for development and use Onboard wiring terminal power supply interface, supporting a wide voltage range of 7V~36V power supply, convenient for industrial power supply use Onboard RTC clock chip, supporting timed tasks On board digital isolation to avoid external signal interference with the control chip Onboard integrated power isolation can provide stable isolation voltage, and the isolation end does not require additional power supply Onboard RS485 transceiver indicator light and CAN working indicator light, convenient for sensing the operating status of the device Rail type protective shell, easy to install and use, with a shell for greater safety Parameter: Model: ESP32-S3-RS485-CAN Microcontroller: ESP32-S3 (default ESP32-S3R8 chip) Wireless communication: 2.4GHz Wi Fi (802.11 b/g/n), Bluetooth 5 (LE) USB interface form: USB Type-C Supply voltage: 5V Function: Device power supply, USB communication, firmware download, etc Isolation type RS485 communication interface: Interface form: terminal block Direction control: Automatic control through the main control hardware flow setting Interface protection: Provides TVS tube protection, surge and electrostatic protection Matching resistor: default NC, reserved 120R matching resistor, can be enabled by jumper cap Isolation type CAN communication interface: Interface form: terminal block Direction control: Hardware automatically identifies and controls the direction of data transmission Interface protection: Provides TVS tube protection, surge and electrostatic protection Matching resistor: default NC, reserved 120R matching resistor, can be enabled by jumper cap PWR power indicator light: connected to USB, red light on when voltage is detected RS485 indicator light: When data is sent from the RS485 interface, the green light will turn on When data is sent back from the RS485 interface, the blue light will illuminate CAN indicator light: Flashing when there is data communication on the CAN interface Terminal power supply range: 7V~36V Shell: Rail type protective shell Size: 91.6 * 23.3 * 58.7 (mm) Package Include: ESP32-S3-RS485-CAN * 1 SH1.0 4PIN connection wire approximately 150mm x 1 Screwdriver x 1


GND
VCC
GND
3V3
IO2
IO1
DC 7~36V
CAN
RS485
PWR
ESP32-S3-RS485-CAN
CAN
RS485
L
H
B-
A+

Powerful AI Computing Capabilities

Based on ESP32-S3, dual-core 240MHz, delivering robust AI computing power.

Supports Wi–Fi + BT 5 (LE)

Supports Wi-Fi + BT 5 (LE) for more stable wireless connections

Isolated RS485/CAN Interface

Isolated RS485/CAN Interface, Safer Industrial Communication

On–board Indicator Lights

On-board RS485 and CAN indicator lights for easy
monitoring of device operational status


ESP32-S3-RS485-CAN
CAN
L
H
RS485
B-
A+
PWR
DC 7-36V
GND
VCC


GND
VCC
GND
3V3
IO2
IO1
DC 7~36V
CAN
RS485
PWR
ESP32-S3-RS485-CAN
CAN
RS485
L
H
B-
A+


GND
VCC
DC 7~36V
CAN
RS485
PWR
ESP32-S3-RS485-CAN
CAN
RS485
L
H
B-
A+


GND
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GND
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IO1
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ESP32-S3-RS485-CAN
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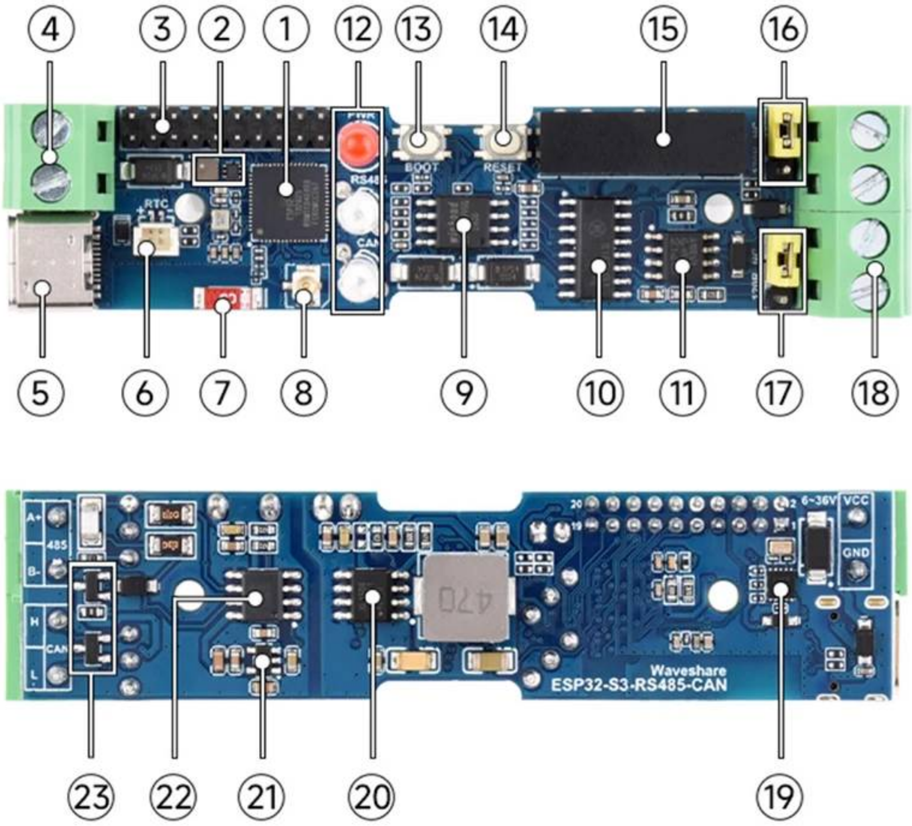
Resource Overview



1、 ESP32-S3

Operating at up to 240MHz, featuring 2.4GHz Wi-Fi and low-power BT

2、 DC-DC Power Module

Power module supporting up to 3.3V 2A output

3、 Pin Header Interface (2.0mm pitch)

Connects to other functional modules via pin headers

4、 Power Terminal Block

Supports wide voltage input range of DC 7V to 36V

5、 USB Type-C Port

Supports device power delivery, firmware download, and USB communication

6、 RTC Battery Connector

Connects to rechargeable RTC battery (SH1.0)

7、 Surface-mount Ceramic Antenna

8、 IPEX Gen1 Socket

Can be switched to external antenna by desoldering resistors

9、 16MB Flash

10、Digital Isolation

Prevents external signal interference from affecting the control chip

11、CAN Conversion Chip

12、Operational Indicator Lights

PWR: Power indicator light
RS485: Green light illuminates when data is transmitted from the RS485 interface;
Blue light illuminates when data is received via the RS485 interface
CAN: CAN indicator light; flashes when data communication occurs on the CAN interface

13、Device BOOT Button

14、Device Reset Button

15、Power Isolation

Provides stable isolated voltage; isolated terminals require no additional power supply

16、RS485 Terminating Resistor

Reserved for 120Ω terminating resistor; enabled via jumper cap

17、CAN Termination Resistor

Pre-installed 120Ω termination resistor, enabled via jumper cap

18、RS485 & CAN Communication Interfaces

Supports connection to external RS485 and CAN devices

19、RTC Clock Chip

PCF85063 Real-Time Clock (RTC)

20、DC-DC Power Supply Chip

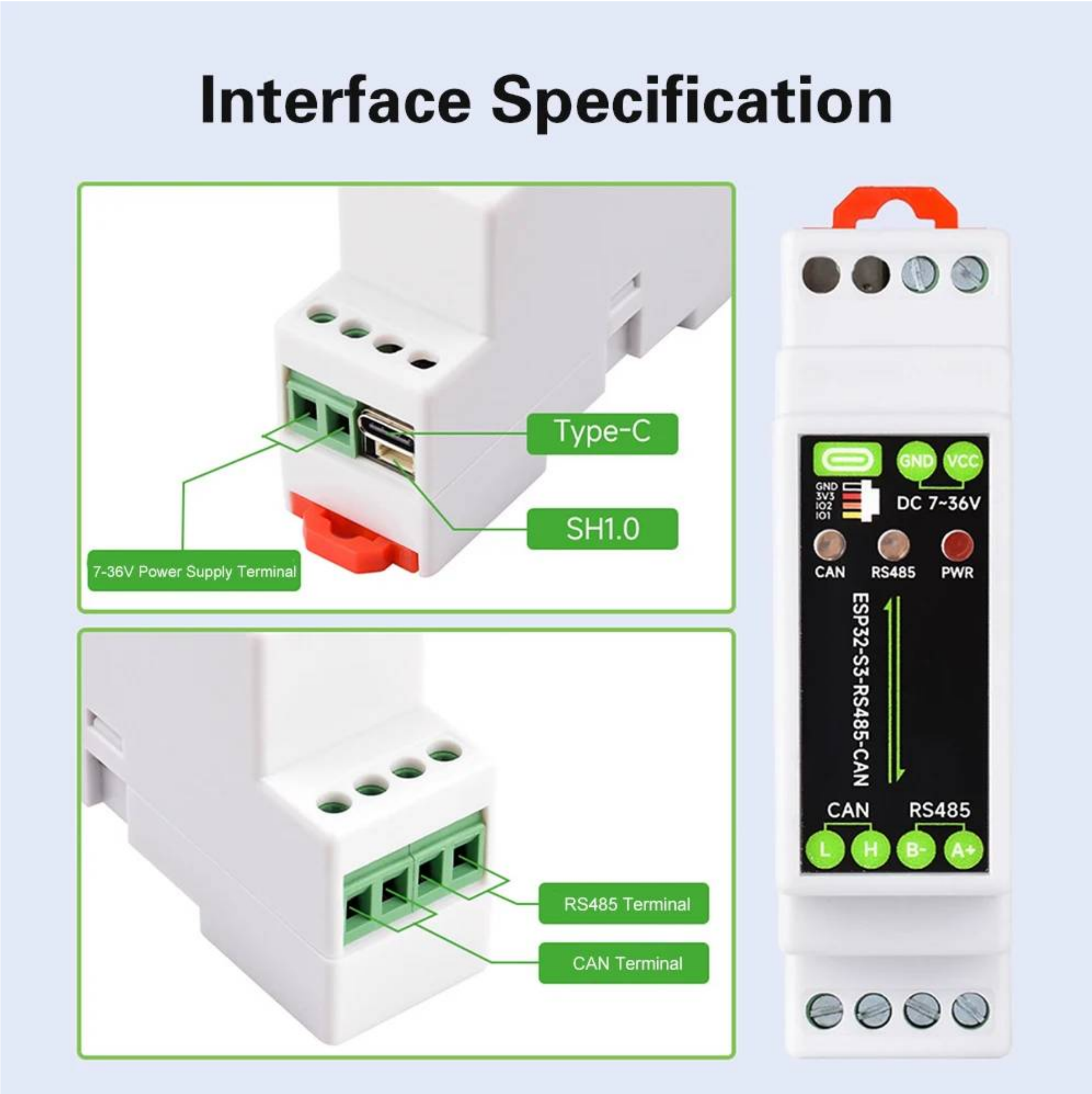
21、ME6217C33M5G

Low Dropout LDO, Current (Max) 800mA

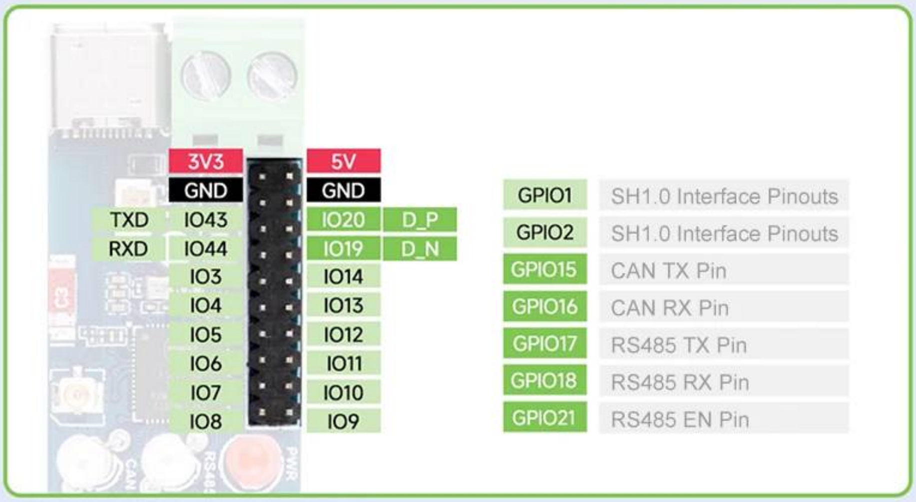
22、RS485 Conversion Chip

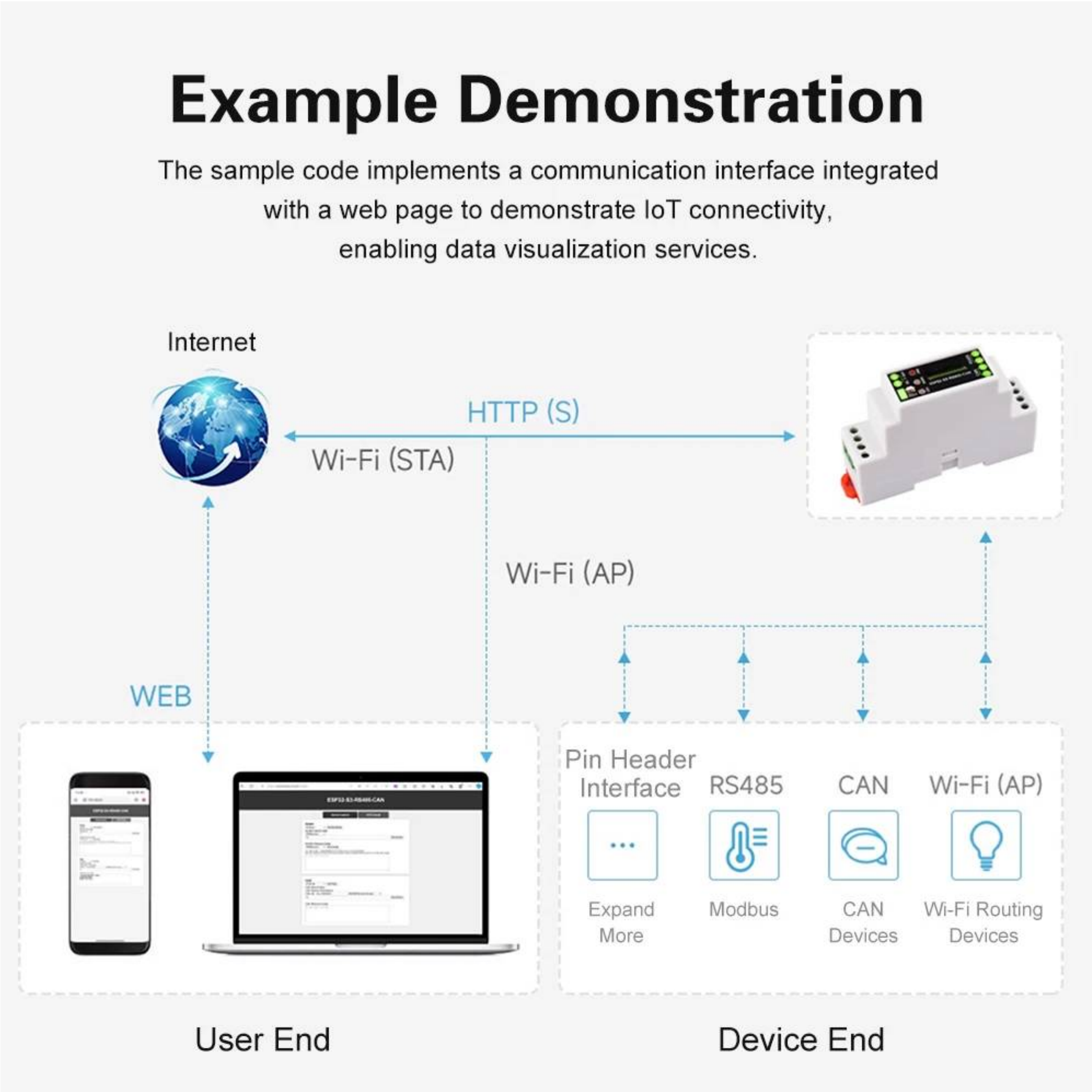
23、On-Board TVS (Transient Voltage Suppressor)

Effectively suppresses surge voltages and transient spikes in the circuit


Interface Specification
Type-C
SH1.0
7-36V Power Supply Terminal
RS485 Terminal
CAN Terminal
GND
VCC
DC 7~36V
GND
3V3
IO2
IO1
CAN
RS485
PWR
ESP32-S3-RS485-CAN
CAN
RS485
L
H
B-
A+

Upper Terminal Description			Lower Terminal Description	
Power supply terminal	Power supply positive terminal: DC 7V to 36V		RS485 Terminal	A+ signal pin
	Negative Power Terminal			
Type-C	Can be used for device power supply, firmware download, and USB communication.			B- signal pin
SH1.0	GND	Ground	CAN Terminal	H signal pin
	3V3	3.3V Power Supply Output		
	GPIO2	Expansion 10		L signal pin
	GPIO1	Expansion IO		



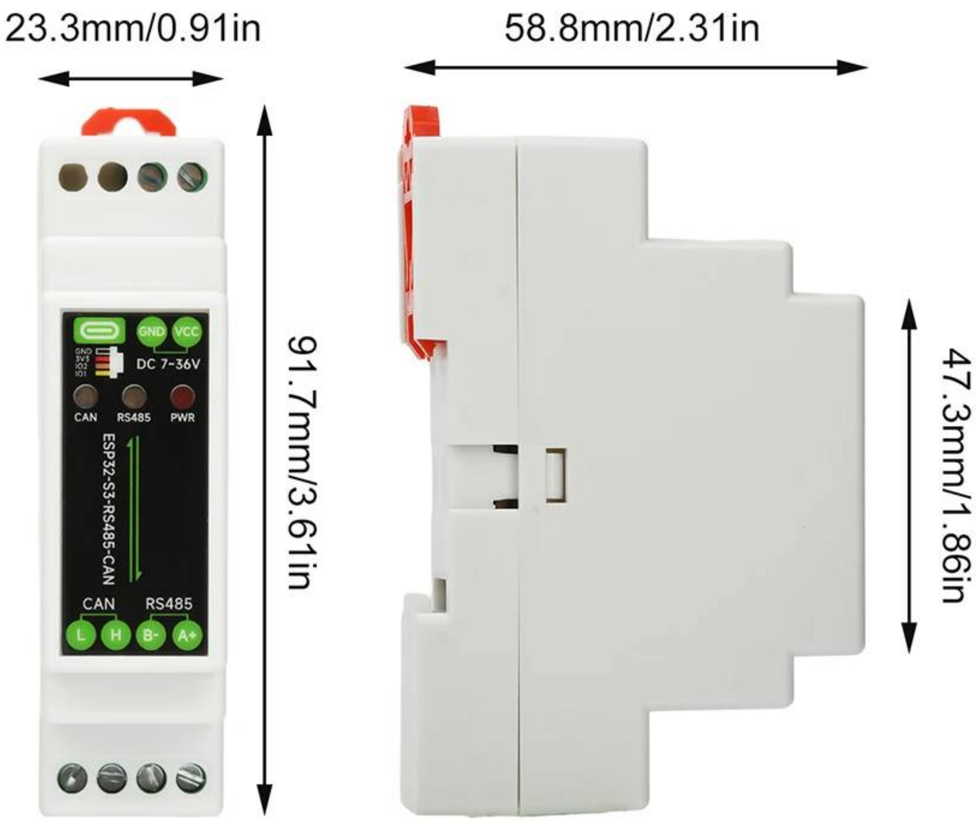

Example Demonstration
The sample code implements a communication interface integrated with a web page to demonstrate IoT connectivity, enabling data visualization services.
Internet
HTTP (S)
Wi-Fi (STA)
Wi-Fi (AP)
WEB
Pin Header Interface
RS485
CAN
Wi-Fi (AP)
Expand More
Modbus
CAN Devices
Wi-Fi Routing Devices
User End
Device End

Product Features

- Based on the ESP32-S3 microcontroller, featuring an integrated Xtensa 32-bit LX7 dual-core processor with a maximum clock speed of 240MHz
- Integrated 2.4GHz Wi-Fi and Bluetooth 5 (LE) dual-mode wireless communication with low power consumption, delivering exceptional RF performance
- On-board isolated RS485 interface for convenient expansion and connection to various RS485 Modbus J industrial expansion modules modules or sensors.
- On-board isolated CAN interface for easy integration with various CAN devices.
- On-board header pins for connecting additional devices.
- On-board USB Type-C port for power supply and download/debugging, facilitating development.
- On-board terminal block power supply interface supporting a wide voltage range of 7V to 36V, suitable for industrial power applications.
- On-board RTC clock chip supports scheduled tasks
- On-board digital isolation prevents external signal interference with control chips
- On-board integrated power isolation provides stable isolated voltage; isolated terminals require no additional power supply
- On-board RS485 transmit/receive indicator lights and CAN operation indicator lights facilitate monitoring device status
- Rail-mount protective enclosure simplifies installation and enhances safety

Product Specifications

Model: ESP32-S3-RS485-CAN
Microcontroller: ESP32-S3 (default chip: ESP32-S3R8)
Wireless Communication: 2.4GHz Wi-Fi (802.11 b/g/n), Bluetooth 5 (LE)
USB Interface Type: USB Type-C
Supply Voltage: 5V
Functions: Device power supply, USB communication, firmware download, etc.
Isolated RS485 Communication Interface:
Interface Type: Terminal Block
Direction Control: Automatically controlled via host hardware flow setting
Interface Protection: Provides TVS diode protection, surge and electrostatic protection
Matching Resistor: Default NC, reserved for 120Ω resistor, enabled via jumper cap
Isolated CAN Communication Interface:
Interface Type: Terminal Block
Direction Control: Hardware automatically detects and controls data transmission direction
Interface Protection: Provides TVS diode protection, surge and ESD protection
Matching Resistor: Default NC, reserved for 120Ω resistor, enabled via jumper cap
PWR Power Indicator: Connects to USB; illuminates red when voltage detected
RS485 Indicators:
Illuminates green when data is transmitted from the RS485 interface
Illuminates blue when data is received back via the RS485 interface
CAN Indicator: Flashes when data communication occurs on the CAN interface
Terminal Supply Voltage Range: 7V ~ 36V
Enclosure: Rail-mounted protective enclosure



23.3mm/0.91in
58.8mm/2.31in
91.7mm/3.61in
47.3mm/1.86in
GND
VCC
DC 7~36V
CAN
RS485
PWR
ESP32-S3-RS485-CAN
CAN
RS485
L
H
B-
A+